1 Bandwidth threshold for k-set agreement

Setting. There are n agents on a directed graph G. Agent i holds an input $x_i \in V$. In one round it broadcasts a single message $s_i(x_i) \in [m]$ to its out-neighbours, and then decides $d_i\big(x_i, (s_j(x_j))_{j \in N^-(i)}\big) \in V$. A protocol solves *$k$-set agreement* if, on every input vector, (validity) every output is some agent's input, and (agreement) at most k distinct values are output. We assume $n \geq k+1$; for $k \geq n$ every agent can output its own input.

Theorem 1 (Lower bound, any topology)**.** *If $|V| \geq m + k$, then no one-round protocol solves k-set agreement on any directed graph G.*

Theorem 2 (Upper bound, complete graph)**.** *If $|V| \leq m+k-1$, then there is a one-round protocol that solves k-set agreement on the complete graph.*

Corollary 3. *On the complete graph with one round, k-set agreement is solvable if and only if $|V| \leq m + k - 1$. For $k = 1$ this is Theorem 8.1.*

Lemma 4 (Product fibres)**.** *For each agent i, let $C_i \subseteq V$ be a fibre of s_i (a set on which s_i is constant) with $|C_i| \geq 2$. On every input vector in $P = C_1 \times \cdots \times C_n$, every agent outputs its own input.*

Proof. On P every message is constant, so agent i's decision is a function $f_i(x_i)$ of its own input alone. Suppose $f_i(c) = v \neq c$ for some $c \in C_i$. Because $|C_j| \geq 2$, we can choose $x_j \in C_j \setminus \{v\}$ for every $j \neq i$. Then the vector with $x_i = c$ and these x_j lies in P and contains no agent holding v, so the output v violates validity. Hence $f_i(c) = c$. □

Proof of Theorem 1. Fix agent i. Let t_i be the number of fibres of s_i with at least two elements, and let U_i be their union. There are at most m fibres in total and at most $m - t_i$ of them are singletons, so

$$|U_i| \;\geq\; |V| - (m - t_i) \;\geq\; k + t_i.$$

In particular $t_i \geq 1$.

Choose fibres greedily for agents $1, \ldots, k+1$. Suppose agents $1, \ldots, i-1$ already have fibres $C_1, \ldots, C_{i-1}$ with distinct representatives $r_1, \ldots, r_{i-1}$. If every non-singleton fibre of s_i were contained in $\{r_1, \ldots, r_{i-1}\}$, then $|U_i| \leq i - 1 \leq k$, which contradicts $|U_i| \geq k + t_i \geq k + 1$. So agent i has a fibre C_i with $|C_i| \geq 2$ that contains a new representative $r_i \notin \{r_1, \ldots, r_{i-1}\}$. For each agent $j > k+1$, take any fibre C_j with $|C_j| \geq 2$, which exists because $t_j \geq 1$.

Now take the input vector with $x_i = r_i$ for $i \leq k+1$ and any $x_j \in C_j$ for $j > k+1$. It lies in $P = \prod_i C_i$, so by Lemma 4 every agent outputs its own input. That gives at least $k+1$ distinct outputs, a contradiction. The argument never uses the edges of G, so it holds for every topology. □

Proof of Theorem 2. Write $V = \{0, \ldots, |V|-1\}$. Let $L = \{0, \ldots, m-2\}$ be the set of *low* values and $H = V \setminus L$ the set of *high* values, so $|H| \leq k$. Each agent sends $s(x) = x$ if $x \in L$ and $s(x) = m-1$ otherwise. Agent i forms the set of low values it knows: its own input if that is low, together with every low value it decodes from a message. If this set is non-empty, agent i outputs its minimum; otherwise it outputs x_i.

Suppose some input is low. On the complete graph every agent knows every low input, either as its own or by decoding, so all agents compute the same non-empty set. They therefore output the same value, and that value is an input. Otherwise all inputs lie in H and each agent outputs its own input. These outputs are valid, and there are at most $|H| \leq k$ distinct ones. (If $m = 1$, then $L = \emptyset$ and $|V| \leq k$.) □

Remark 5. *Both directions use only fibres of the message maps; no homology enters at $R = 1$, for consensus or for k-set agreement. Below the threshold, the topology does matter. For example, with $n = 3$, $k = 2$, $m = 2$ and $|V| = 3$, the task is solvable on the complete graph and on the directed ring but impossible on the chain and on the* in-star *(leaves send to the hub and hear nothing), by exhaustive search (report 03, confirmed by SAT in Section 1.1). On the* bidirectional *star it is solvable: see Observation 10. Report 03's one-round "star" is the in-star: its recorded reachability sets are $\{0, 1, 2\}$ for the hub and $\{i\}$ for each leaf i. If agents may send different messages on different edges, the same proof gives the lower bound with m replaced by $m^{d_i^+}$, the size of agent i's combined out-message alphabet.*

1.1 Other topologies and two rounds: exhaustive computations

Method. For a digraph G, $R \geq 1$ rounds, m, k and $|V|$ we encode the existence of a protocol as a CNF formula and decide it with the complete SAT solver CaDiCaL 1.5.3 (via PySAT). In round r agent j broadcasts $s_j^r(x_j, h_j) \in [m]$, where h_j is everything j received in rounds $1, \ldots, r-1$, so later messages are fully adaptive. After round R agent i decides $d_i(x_i, h_i) \in V$. The variables are the complete tables of all s_j^r and d_i. Auxiliary variables record which symbol each agent sends on each input vector, and which values are output on each input vector. They appear only as heads of Horn

implications, so the formula is satisfiable if and only if a protocol exists. The only symmetry breaking is that, for each sending agent and each history, the symbol sent on input 0 is fixed to 0. This is sound because receivers can undo any relabelling: in round 1 always, and in later rounds whenever every receiver knows the sender's history, which always holds on the complete graph. An UNSAT answer is therefore an exhaustive impossibility proof over the whole protocol space. Every SAT answer was checked again by extracting the protocol and simulating it on all $|V|^n$ inputs. On the complete graph the encoder reproduces the original one-round script, and it agrees with all 46 consensus and k-set cells of report 03 when report 03's one-round star is read as the in-star (see the remark above). Raw data and the scripts are in `results/` and `sat/`.

Topologies. *complete*; *star*: hub 0 and leaves, with edges in both directions; *in-star*: leaves $\to$ hub only; *out-star*: hub $\to$ leaves only; *chain*: $0 \to 1 \to \cdots \to n-1$; *ring*: directed cycle $i-1 \to i$.

1.1.1 One round

For each topology, $n \in \{3,4\}$, $k \in \{1,\ldots,n-1\}$ and $m \in \{2,3\}$, we decided solvability for *every* $|V| \in \{k+1,\ldots,m+k\}$. Values $|V| \le k$ are trivially solvable, and $|V| = m+k$ is impossible by Theorem 1; the SAT runs confirm this as well. That is 150 SAT instances. The largest protocol space has about 10^{516} protocols (complete graph, $n=4$, $k=3$, $m=3$, $|V|=6$). In every case the solvable values form an initial segment $\{1,\ldots,T\}$, as they must, since restricting the inputs preserves solvability. Table 1 lists the threshold T.

Observation 6 (Table 1)**.** *On the complete graph every threshold equals $m+k-1$, in agreement with Corollary 3. Every other topology attains $m+k-1$ only in some cells. The star attains it for $k \in \{1, n-1\}$; the ring for $k = n-1$; the out-star only for $k = 1$; the chain and the in-star never. For $n = 4$ and $k = 2$, every non-complete topology falls strictly below $m+k-1$.*

Some entries of Table 1 are explained by the following two results, which we prove.

Lemma 7 (Deaf agents)**.** *Let $|V| \ge 2$ and $n \ge 2$. If agent i has no in-neighbours, then in every one-round protocol that satisfies validity, agent i outputs its own input.*

n	k	m	$m+k-1$	complete	star	in-star	out-star	chain	ring
3	1	2	2	2	2	1	2	1	1
3	1	3	3	3	3	1	3	1	1
3	2	2	3	3	3	2	2	2	3
3	2	3	4	4	4	3	3	3	4
4	1	2	2	2	2	1	2	1	1
4	1	3	3	3	3	1	3	1	1
4	2	2	3	3	2	2	2	2	2
4	2	3	4	4	3	2	3	3	3
4	3	2	4	4	4	3	3	3	4
4	3	3	5	5	5	4	3	4	5

Table 1: Exact one-round thresholds: k-set agreement is solvable iff $|V| \leq T$. Every entry comes from exhaustive SAT over all $|V| \leq m + k$.

Proof. Agent i's decision is a function $f(x_i)$ of its own input. If $f(c) = v \neq c$, give every other agent an input in $V \setminus \{v\}$, which is non-empty. Then no agent holds v, so outputting v violates validity. □

Corollary 8. *If G has at least $k+1$ agents of in-degree 0, then the one-round threshold for k-set agreement is exactly k, for every m.*

Proof. If $|V| \leq k$, every agent outputs its own input. If $|V| \geq k + 1$, give $k + 1$ deaf agents distinct inputs. By Lemma 7 they output $k + 1$ distinct values. □

This explains the in-star entries with $k < n - 1$.

Theorem 9 (One-round consensus on a digraph)**.** *Let $n \geq 2$ and $|V| \geq 2$. One-round consensus is solvable on G if and only if (i) some agent u is an in-neighbour of every other agent, and (ii) $|V| \leq m$. So the threshold is m if G has such a "universal broadcaster", and 1 otherwise.*

Proof. ($\Leftarrow$) Agent u sends x_u, which is possible since $|V| \leq m$, and every agent outputs x_u.

($\Rightarrow$) Condition (ii) is Theorem 1 with $k = 1$. For (i), let $S_i = \{i\} \cup N^-(i)$. Agent i's output depends only on x_{S_i}. Since all agents agree on every input, there is one function $D : V^n \to V$ with $D(x) = d_i(\dots)$ for every i. So, for every i, $D(x)$ depends only on x_{S_i}.

Next we show that if a function on a product set depends only on the coordinates in A, and also only on those in B, then it depends only on those

in $A \cap B$. Let x, y agree on $A \cap B$. Let z agree with x on A and with y off A. Then $D(z) = D(x)$. Also z and y agree on $(A \cap B) \cup A^c \supseteq B$, so $D(z) = D(y)$.

By induction, D depends only on x_S, where $S = \bigcap_i S_i$. If $S = \emptyset$, then D is constant. But validity forces $D(0, \ldots, 0) = 0$ and $D(1, \ldots, 1) = 1$, a contradiction. So S contains some u, and $u \in S_i$ for all i means $u \in N^-(i)$ for all $i \neq u$. □

The $k = 1$ rows of Table 1 agree with Theorem 9. The complete graph, the star and the out-star have a universal broadcaster (any agent, or the hub). The in-star, the chain and the ring with $n \geq 3$ do not.

Observation 10 (Explicit protocol)**.** *On the bidirectional star with* $n = 3$*,* $k = 2$*,* $m = 2$ *and* $V = \{0, 1, 2\}$*, the following protocol solves* 2*-set agreement. It was extracted from the SAT model and checked on all* 27 *inputs. Every agent sends the bit* $[x = 2]$*. The hub outputs* 2 *if it holds or hears a* 2*, and otherwise outputs its own input. A leaf with input* 0 *outputs* 2 *if the hub's bit is* 1 *and* 0 *otherwise. Any other leaf outputs its own input.*

The remaining patterns in Table 1 are unproved.

Conjecture 11 (Set agreement, $k = n - 1$)**.** *For* $k = n - 1$*, the one-round threshold on a digraph* G *is* $m + n - 2$ *(that is, it meets the bound of Theorem 1) if and only if every agent has an in-neighbour. If some agent has in-degree* 0*, the threshold is at most* $m + n - 3$*.*

Evidence: the $k = n - 1$ rows of Table 1, i.e. six topologies, $n \in \{3, 4\}$ and $m \in \{2, 3\}$, giving 24 cells. The complete graph, the star and the ring have no deaf agent and reach $m + n - 2$. The chain, the in-star and the out-star each have a deaf agent and stay at or below $m + n - 3$. The chain and the in-star equal $m + n - 3$ in all four cells; the out-star equals it in three of the four.

Conjecture 12 (Out-star)**.** *On the out-star with* $n \geq 3$ *agents, the one-round threshold for* k*-set agreement is* $\max(m, k)$*.*

Evidence: 10 cells, namely $n \in \{3, 4\}$, $1 \leq k \leq n - 1$ and $m \in \{2, 3\}$. The case $k = 1$ is Theorem 9.

Conjecture 13 (Chain)**.** *On the directed chain with* $n \geq 3$ *agents and* $2 \leq k \leq n - 1$*, the one-round threshold is* $m + k - 2$*.*

Evidence: 6 cells, namely $(n, k) \in \{(3, 2), (4, 2), (4, 3)\}$ and $m \in \{2, 3\}$. For $k = 1$ the threshold is 1 by Theorem 9.

1.1.2 Two rounds

We now allow $R = 2$ with fully adaptive round-2 messages and fix the complete graph with $n = 3$. For consensus the threshold is exactly m^2. Our search found no advantage from adaptivity. This is in fact a theorem for every R, every $n \geq 2$, and every topology.

Theorem 14 (R-round consensus)**.** *Let $n \geq 2$ and $R \geq 1$.*

(a) On every digraph, if $|V| \geq m^R + 1$, then no R-round protocol (adaptive or not) solves consensus.

(b) If some agent u is an in-neighbour of every other agent (for example, on the complete graph), then consensus is solvable in R rounds whenever $|V| \leq m^R$.

Hence the R-round consensus threshold on the complete graph is exactly m^R.

Proof. (b) Agent u sends the R base-m digits of x_u, one per round, and every agent outputs x_u.

(a) We build sets $C_i^{(0)} \supseteq C_i^{(1)} \supseteq \cdots \supseteq C_i^{(R)}$ for each agent i. They satisfy $|C_i^{(r)}| \geq m^{R-r} + 1$, and on $P_r = \prod_j C_j^{(r)}$ every message of rounds $1, \ldots, r$ is constant. Start with $C_i^{(0)} = V$. Suppose the sets for $r-1$ are built. On P_{r-1} all messages before round r are constant, so each agent's history before round r is constant. Hence on $C_i^{(r-1)}$ the round-r message of agent i is a function of x_i alone, with at most m values. Because $|C_i^{(r-1)}| \geq m^{R-r+1} + 1 > m \cdot m^{R-r}$, one of its fibres $C_i^{(r)}$ has at least $m^{R-r} + 1$ elements. Restricting each coordinate to $C_j^{(r)}$ keeps all earlier messages constant, which completes the induction.

On P_R every message in every round is constant, and $|C_i^{(R)}| \geq 2$ for all i. The argument of Lemma 4 applies unchanged: each decision is a function of the agent's own input, and validity forces it to be the identity. Choose $x_1 \in C_1^{(R)}$ and $x_2 \in C_2^{(R)} \setminus \{x_1\}$, and the other coordinates arbitrarily in P_R. Then agents 1 and 2 output different values, so consensus fails. □

Remark 15. *The proof of Theorem 9 carries over verbatim to R rounds once S_i is replaced by the set of agents that have a directed path of length at most R to i. So R-round consensus with $|V| \geq 2$ needs an agent that reaches everyone within R hops. When that agent reaches others only through relays, Theorem 14(b) does not apply, and the threshold may be below m^R.*

Proposition 16 (R-round k-set, complete graph, upper bound)**.** *On the complete graph, k-set agreement is solvable in R rounds whenever $|V| \leq m^R + k - 1$.*

Proof. Run the protocol of Theorem 2 with alphabet size $M = m^R$. Each agent transmits its symbol $s(x) \in [M]$ non-adaptively, as R base-m digits over R rounds. □

task	m	$\|V\|$ checked	threshold	$m^2 + k - 1$	largest protocol space
consensus ($k = 1$)	2	$2, \ldots, 5$	4	4	10^{190}
consensus ($k = 1$)	3	$2, \ldots, 9$	≥ 9 *	9	10^{2216}
2-set ($k = 2$)	2	$3, \ldots, 6$	5	5	10^{251}

Table 2: Two rounds, complete graph, $n = 3$, fully adaptive round-2 messages. Exhaustive SAT for every listed $|V|$. *The SAT instance for $|V| = 10$ ($\approx 7 \cdot 10^4$ variables, $2.6 \cdot 10^6$ clauses, protocol space $\approx 10^{2573}$) did not finish. We ran CaDiCaL and Glucose for more than an hour each, and also a cube-and-conquer split over agent 0's round-1 map, which did not finish either. Impossibility at $|V| = 10$, and hence the exact threshold 9, comes from Theorem 14(a), not from the computation.

Observation 17 (Table 2)**.** *For $n = 3$ on the complete graph with $R = 2$, the consensus threshold is $4 = 2^2$ for $m = 2$. The search covered every $|V| \leq 5$, so this independently confirms both parts of Theorem 14. For $m = 3$ the search shows that consensus is solvable for every $|V| \leq 9 = 3^2$, which confirms part (b). The impossibility side at $|V| = 10$ rests on the proof of part (a) alone, because the SAT run did not finish (see Table 2). For 2-set agreement with $m = 2$ the threshold is $5 = m^2 + k - 1$, which matches Proposition 16.*

Conjecture 18 (R-round k-set agreement)**.** *On the complete graph, the R-round threshold for k-set agreement is exactly $m^R + k - 1$. Equivalently, adaptivity gives no advantage over a single round with alphabet $[m]^R$.*

Evidence: Theorem 14 ($k = 1$, all R), Corollary 3 ($R = 1$, all k), and the cell $n = 3$, $k = 2$, $m = 2$, $R = 2$ in Table 2. The proof of Theorem 1 does not extend directly. Its greedy choice of fibres with distinct representatives uses fibres of s_i that do not depend on the other agents. With adaptivity, agent i's round-2 fibres depend on the round-1 fibres chosen for everyone else.